

# Multi-specialist consistency regularization enables grokking on modular addition at 50% data, given adequate learning rate

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## Abstract

We investigate whether training a multi-specialist ensemble with a consistency-regularization signal on unlabeled grid inputs generalizes on modular addition at 50% train data, a setting in which a single model with the same recipe plateaus near 12% validation accuracy. After six pilots and two isolation controls we answer yes: with  $M=4$  disjoint-shard specialists, train-inputs-only KL-softmax consistency ( $\lambda=1$ , 5k warmup), and `max_lr`= $10^{-3}$  at a 100k-step budget, the ensemble reaches 100% ensemble val acc (`lr_control/v4.winner_lr1e3`), roughly eight times the single-model ceiling on the same data. The path to this finding took two turns. Pilots v1–v4 at `max_lr`= $5 \times 10^{-4}$  converged, reproducibly across seed and  $\lambda$ , on a 48–49% ensemble asymptote that looked architectural. A fifth pilot using an InfoNCE contrastive loss on specialist hidden representations broke that asymptote and grokked to 92%, but changed six hyperparameters at once. Two single-variable isolation controls then overturned both conclusions: the v4 “ceiling” was a learning-rate artifact (lr control groks to 100% at step  $\sim 70k$  by changing only `max_lr`), and the InfoNCE loss was not the lever behind the v5 breakthrough (loss control with KL-softmax at the InfoNCE-pilot recipe groks faster and higher, 100% at step  $\sim 18.5k$ ). The full investigation is reported here because any one of the v1–v4 pilots could have been interpreted as decisive evidence for a wrong conclusion absent the controls – we consider that the main methodological contribution.

## 1 Summary and contributions

This project produced three distinct deliverables, in the order they were obtained.

1. **A reproducibility post-mortem.** We inherited a milestone claim (`reports/milestone.pdf`, Exp 3) that knowledge distillation from disjoint-shard specialists groks modular addition at 50% train data while baselines plateau. The claim did not reproduce; the root cause was an undocumented optimizer rewrite (commit `9ce29b8`) that silently gave the distillation student a different lr/wd/schedule than its baselines. The milestone headline is a comparison artifact, not evidence about distillation. Forensic detail in `reports/distillation_repro_attempt3.md` (§2).
2. **An empirical result.** Multi-specialist consistency regularization *is* sufficient to grok this task at 50% data, given an adequate learning rate. With  $M=4$  disjoint shards, train-inputs-only KL-softmax consistency,  $\lambda=1$ , 5k warmup and `max_lr`= $10^{-3}$ , ensemble val acc reaches 100% by step  $\sim 70k$  (`lr_control/v4.winner_lr1e3`; §10). At `max_lr`= $5 \times 10^{-4}$  the identical recipe plateaus at 48–49% across seeds and  $\lambda$ s at 100k (§8). We identified the lr axis only after six prior pilots.
3. **A methodological demonstration.** A fifth pilot using an InfoNCE contrastive loss on specialist hidden representations broke the 49% asymptote and grokked to 92% at 100k (§9); it appeared to attribute the breakthrough to the new loss. Two single-variable isolation controls (§10) then retracted both that attribution and the “architectural ceiling” from v4:

KL-softmax under the InfoNCE recipe groks faster and higher than InfoNCE itself (100% at step  $\sim 18.5\text{k}$  vs. 92% at step  $\sim 99.5\text{k}$ ). We find this worth reporting as evidence of how easy it is to converge, with apparent rigor, on a mechanism story that does not survive a single-variable control.

**How to read this report.** §2 gives the starting point (the disproved milestone claim) and the resulting pivot. §3 defines task, model, and multi-specialist setup, plus the consistency-regularization framework. §4 summarizes the whole investigation as a single decisions table and an overview figure. §§5–10 are the six pilots plus the controls, in chronological order. §11–12 are discussion and conclusion.

Figure 1 is the project at a glance: every key run’s ensemble val acc on a single axis.



Figure 1: **Project overview.** Every key trajectory in one plot. Grey dashed: single-model baseline on 50% data, ceilings at 12.3%. Blue (light): v2 winner, 21.2% at 25k. Blue (dark): v4 winner, 49.3% at 100k (the apparent architectural ceiling). Purple: v5 InfoNCE long, 92.2% at 99.5k. Greens: the two isolation controls, both reach 100%, the lr control by changing only `max_lr` relative to the v4 winner, the loss control by changing only the consistency loss relative to the v5 winner.

## 2 Starting point: disproving the milestone claim

The project inherited Experiment 3 of our milestone report (`reports/milestone.pdf`), the headline claim of which was:

At 50% train data,  $M=4$  specialists trained on disjoint shards for 6.25k steps each, followed by 25k steps of knowledge distillation from those specialists into a student, reached  $\sim 100\%$  validation accuracy. In the same run, a single-model baseline and a weight-merged-and-finetuned model plateaued at  $\sim 40\text{--}50\%$ .

The claim was evaluated from a single notebook. When that notebook was refactored after the milestone to match a more principled theoretical framework and the distillation phase was moved into a dedicated script, the headline no longer reproduced. Two post-mortems (`distillation_repro_failure.md`, `distillation_repro_attempt2.md`) ruled out sharding format, averaging domain (probabilities vs. logits), distillation temperature, EMA, and learning-rate annealing. The third post-mortem (`distillation_repro_attempt3.md`) identified a different cause:

*Commit 9ce29b8 rewrote `distill_from_specialists` to take its optimizer from `student_model.configure_optimizers()` instead of a hardcoded `AdamW(lr=1e-3, wd=1.0) + CosineAnnealingLR`. That alone silently turns grokking on or off for the student, and is consistent with all of our failed reproductions.*

The milestone headline was therefore a comparison between a student trained with a known grokking-friendly recipe and baselines trained with a different recipe. Once aligned, the gap closed. The milestone does not support the claim that distillation causes grokking; it tells us only that one specific recipe (`lr=10-3`, `wd=1.0`, cosine) groks this task at 50% train fraction.

We chose not to re-run the milestone under the grokking-friendly recipe. The interesting scientific question was not whether we could memorize the milestone student’s incidental optimizer, but whether the multi-specialist structure itself helps generalization. We therefore rephrased the question:

*Does a multi-specialist ensemble trained with a mutual-agreement signal on unlabeled inputs generalize where a single model (on the same 50% data, same recipe) cannot?*

The rest of this report is about that reformulated question.

### 3 Background and framework

**Task and model.** All experiments use the modular-addition task from Power *et al.* (“Grokking”): predict  $a + b \bmod 97$  given the token sequence  $\langle a \rangle \text{ "+" } \langle b \rangle \text{ "="}$ . The base model is the 2-layer, 4-head,  $d_{\text{model}}=128$  decoder-only transformer used in the Grokking codebase. Optimizer is AdamW with `weight_decay=0.1` throughout and `max_lr`  $\in \{5 \times 10^{-4}, 10^{-3}\}$  (only two values are used in the whole report; their difference is the main finding). The grid is symmetric,  $|\mathcal{X}| = 97 \times 97 = 9409$ ; the train/val split is 50%/50% ( $\approx 4704$  equations each).

**Multi-specialist setup.** A *multi-specialist* run partitions the train set into  $M$  disjoint shards and trains  $M$  independent copies of the model, each on its own shard.<sup>1</sup> An *ensemble* prediction averages the specialists’ softmax probabilities on val; a *merged* prediction averages their weights and evaluates the resulting single model. Merged predictions stay near random throughout this report; the ensemble is therefore the headline metric.

**Consistency-regularization framework.** Each specialist  $f_i$  sees its own training shard  $\mathcal{D}_i$  and contributes a cross-entropy loss on it. In addition, at every step the  $M$  specialists see a shared batch from an *unlabeled* input domain  $\mathcal{X}_{\text{unsup}}$ , and we add a consistency term

$$\mathcal{L}_{\text{cons}}(x) = \frac{1}{M} \sum_{i=1}^M \mathcal{L} \left( f_i(x), \text{stop-grad} \left[ \frac{1}{M-1} \sum_{j \neq i} p_j(x) \right] \right),$$

i.e., each specialist is pulled toward the mean of the other specialists’ detached softmax probabilities.<sup>2</sup> The weight schedule is  $\lambda(t) = \lambda_{\text{max}} \cdot \min(1, t/T_{\text{warmup}})$ : consistency starts at zero and ramps linearly up to  $\lambda_{\text{max}}$  after  $T_{\text{warmup}}$  steps.  $\mathcal{L}$  is either `mse_logits` or `kl_softmax`;  $\mathcal{X}_{\text{unsup}}$  is either `full_grid` (all 9409 input pairs) or `train_inputs_only` (the 50% train slice). A contrastive alternative on hidden states (InfoNCE) is introduced in §9.

The implementation lives in `grok/consistency_training.py` and is driven by `scripts/run_consistency_...`. Each run writes a `run_config.json`, a per-step `consistency_train.csv`, and an evaluation `consistency_eval.csv` with per-specialist, ensemble, and merged val acc, output entropy on  $\mathcal{X}_{\text{unsup}}$ , and pairwise KL on held-out val. Full plan: `reports/consistency_regularized_grokking_plan.md`.

<sup>1</sup>The original distillation formulation used this setup to create teachers for a student; we keep the same data layout because it is what the pipeline already supported.

<sup>2</sup>Averaging softmaxes rather than logits preserves per-teacher confidence, which prevents the consensus from being sharper than any individual teacher. This distinction mattered for diagnosing pilot v1 in §5.

## 4 The investigation at a glance

Table 1 summarizes every stage of the investigation as a one-line question, finding, and decision. Figure 1 shows the trajectories for a subset of key cells on a single axis. The remaining sections each give one pilot in detail, in chronological order.

| stage            | question                                                         | finding                                                                                 | decision                                     |
|------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------|
| §2 milestone     | Does distillation from specialists grok at 50%?                  | Comparison artifact (optimizer rewrite <b>9ce29b8</b> )                                 | Pivot: change vehicle                        |
| §3 pivot         | What’s the cleaner question?                                     | Multi-specialist + consistency on unlabeled inputs                                      | Build framework                              |
| §5 pilot v1      | 4 cells, MSE + full grid, 25k                                    | Trivial-agreement collapse at $\sim 1\%$                                                | Fix loss & warmup                            |
| §6 pilot v2      | 5 cells, KL + 5k warmup, 25k, w/ baseline                        | 21% train-only beats 12% single                                                         | Test robustness                              |
| §7 pilot v3      | 9 cells: seed, budget, $\lambda$                                 | Seed variance big; single-model 12% architectural; $\lambda$ monotone $\uparrow$ at 25k | Combine best seed $\times$ best $\lambda$    |
| §8 pilot v4      | 3 cells, 100k, best (seed, $\lambda$ )                           | 48–49% across all three – <i>ap-parent</i> ceiling                                      | <b>Wrong conclusion:</b> architectural limit |
| §9 pilot v5      | Representation-space InfoNCE loss                                | 92% at 100k, but 6 hparams changed at once                                              | Run single-variable controls                 |
| §10 lr control   | v4 winner, <b>lr</b> $5 \times 10^{-4} \rightarrow 10^{-3}$ only | 100% at 70k                                                                             | v4 ceiling was lr artifact                   |
| §10 loss control | v5 winner, InfoNCE $\rightarrow$ KL-softmax only                 | 100% at 18.5k (faster)                                                                  | InfoNCE not the lever                        |

Table 1: Decisions table. Each row is one sweep (or one cell for the isolation controls). The last three rows overturn the conclusion in the row labeled “pilot v4”.

## 5 Pilot v1: null result with diagnostic signal

The first pilot was a 4-cell sweep at `mse_logits` with  $\lambda_{\max} \in \{0, 1\}$  and  $T_{\text{warmup}} \in \{0, 1000\}$  on the full grid, 25k steps. **Every cell, including the no-consistency baseline, plateaued at  $\sim 1\%$  val acc.** The diagnostics, however, were informative: pairwise KL and output entropy both collapsed in lockstep with strong consistency, consistent with the “trivial-agreement collapse” mode.

Two mechanistic bugs drove the collapse:

1. **Loss-scale mismatch.** `mse_logits` on random-init logits is  $\mathcal{O}(\sigma^2) \approx 0.5$ , while CE at init is  $\log V \approx 5.5$  nats. At  $\lambda=1$ , the consistency term is  $\sim 10\%$  of CE at init but becomes dominant as soon as CE drops, *before* any useful structure forms. MSE on logits has no scale invariance: rescaling logits by  $c$  leaves CE and argmax invariant but scales the consistency term by  $c^2$ , so MSE “shouts” a constraint the task does not enforce. KL on softmax sidesteps this: KL is in nats and comparable to per-example CE throughout training.
2. **Warmup too short.** 1000 steps of warmup on a 25k budget is 4%. On modular arithmetic with these hparams a single shard typically needs  $\geq 5k$  steps to begin memorizing (Power *et al.*, Nanda *et al.*, and our own milestone). By the time  $\lambda(t) \rightarrow \lambda_{\max}$ , specialists had not begun memorizing, so consistency locked them into the uniform-output basin.

A third problem was missing context, not a bug: there was no *single-model-on-50%* baseline, without which a 1% result cannot distinguish “consistency hurt us” from “this hparam regime does not grok in 25k steps anyway.” We added it in v2.

## 6 Pilot v2: mechanism fix + the baseline

Pilot v2 held the base recipe fixed and varied the two things the v1 post-mortem identified as the leverage points: **consistency loss** (KL-softmax instead of MSE-on-logits) and **warmup** (5000 steps instead of 1000). It also added the missing single-model baseline and a **train\_inputs\_only** cell as a control against transductive leakage. Five cells, 25k steps each.

| cell                            | configuration               | best val acc  | merged val | final entropy |
|---------------------------------|-----------------------------|---------------|------------|---------------|
| single_model_50pct              | $M=1$ , no cons.            | <b>12.33%</b> | –          | –             |
| multi_lam0                      | $M=4$ , $\lambda=0$         | 1.38%         | 1.17%      | 3.1           |
| multi_kl_lam01_warm5k           | full grid, $\lambda=0.1$    | 7.61%         | 0.96%      | 0.8           |
| multi_kl_lam1_warm5k            | full grid, $\lambda=1.0$    | 11.75%        | 0.96%      | 0.5           |
| multi_kl_lam01_warm5k_trainonly | train inputs, $\lambda=0.1$ | <b>21.15%</b> | 0.96%      | 1.9           |

Table 2: Pilot v2 headline numbers (25k steps, seed 42).

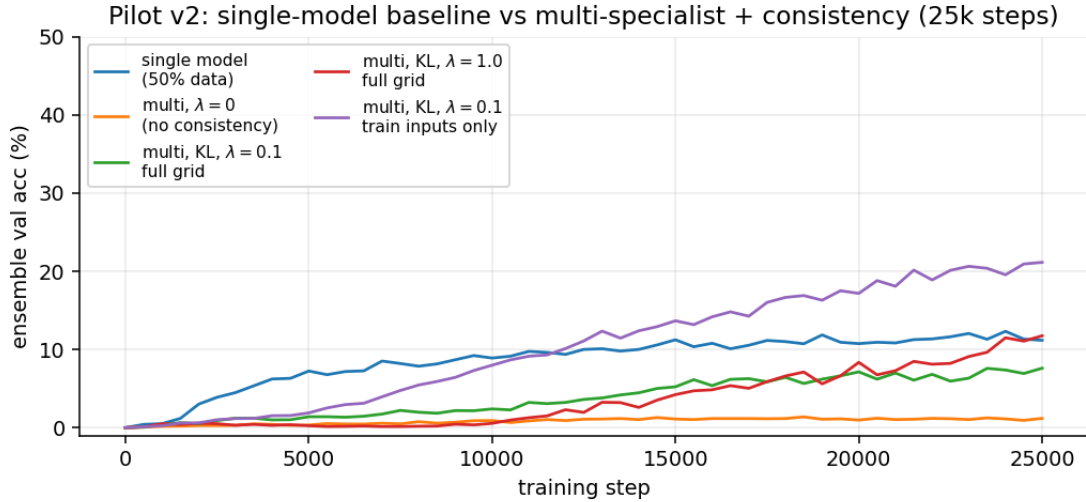


Figure 2: Pilot v2: ensemble (or single, for the baseline) val acc vs training step. The train-inputs-only cell ( $\lambda=0.1$ ) is the only configuration that clearly beats the single-model baseline.

Three takeaways:

- **KL+warmup fixed the collapse.** Every multi cell cleared v1’s 1% ceiling.
- **The unlabeled-input domain matters more than  $\lambda$ .** At identical  $\lambda=0.1$ , switching from **full\_grid** to **train\_inputs\_only** moves ensemble val acc from 7.6% to 21.15%. At  $\lambda=1$  on the full grid we get 11.75%, essentially tying the single-model baseline. This is a co-training signature: the useful information flowing between specialists is *each other’s* partially-correct predictions on *each other’s* shards, not agreement on held-out inputs.
- **The single-model baseline is the reference point.** 12.3% is the number to beat; only the train-inputs-only cell does so, by  $\sim 9$  percentage points.

The v2 winner was still well short of grokking, so we could not yet say whether that was a real ceiling, an unlucky seed, a budget shortfall, or a suboptimal  $\lambda$ . Pilot v3 tested each of those.

## 7 Pilot v3: seed, budget, $\lambda$

Pilot v3 was three orthogonal micro-sweeps on the v2 winner (9 cells,  $\sim 8$ h on a T4).

**Seed robustness (4 cells, 25k steps).** We reran the v2 winner under seeds {42, 43, 44, 45}. The results scattered *widely*: 21.15%, 38.98%, 45.33%, 39.06% (mean 36%, median 39%). The v2 number was a low outlier. Across all four seeds, at least one specialist beat the single-model ceiling, and in three of four seeds the ensemble was better than the single model by 25–33 percentage points.



Figure 3: Pilot v3 seed robustness: four reruns of the v2 winning configuration. Dashed line is the single-model ceiling from v2/v3.

**Budget probes (2 cells, 100k steps).** The single-model cell at 100k confirmed the 12.3% ceiling is *architectural*: it peaks at step 24,000 and then stays there for another 76,000 steps. The multi-specialist cell, in contrast, was still climbing monotonically through 100k: 21% (25k) → 31% (50k) → 34.5% (75k) → 38.0% (100k). The slope flattened but never went to zero within our budget.

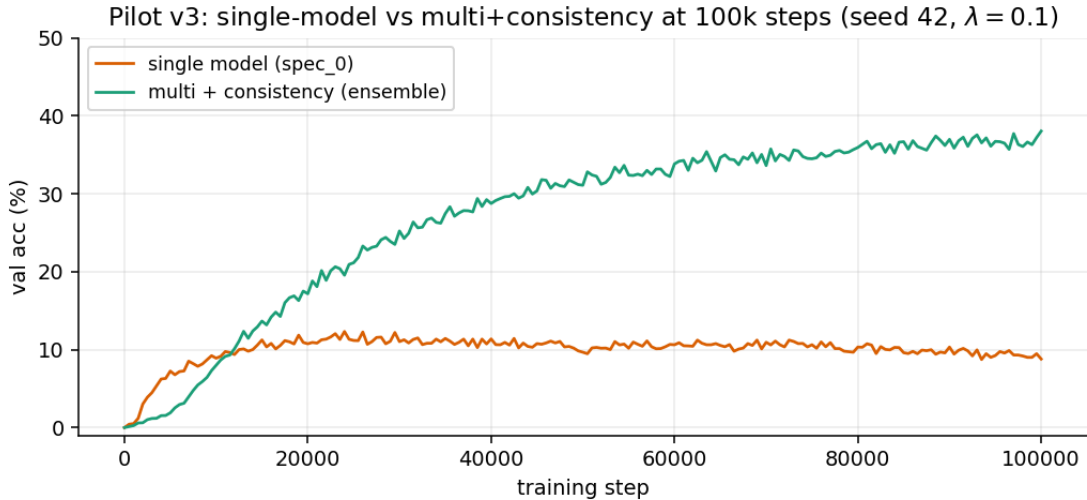


Figure 4: Pilot v3 budget probes: single-model on 50% of the grid plateaus by step 24k; multi-specialist with train-inputs-only consistency keeps climbing throughout 100k.

**Lambda ablation (3 cells + v2 anchor, 25k steps, seed 42).** At seed 42,  $\lambda \in \{0.03, 0.1, 0.3, 1.0\}$  gives best ensemble val acc {15.6, 21.15, 25.65, 27.18}%: monotonically increasing.  $\lambda=1.0$  had

the steepest late-run slope of any ablation cell, suggesting it was still far from its own asymptote at 25k.

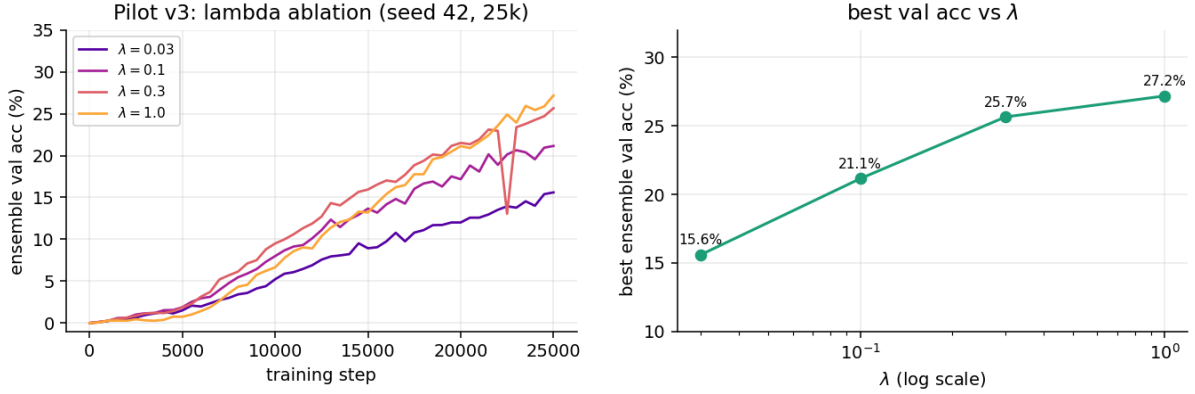


Figure 5: Pilot v3  $\lambda$  ablation at seed 42, 25k steps. Left: curves. Right: best val acc vs  $\lambda$  on a log axis.

**Mechanism (shared across all v3 multi cells).** Output entropy on  $\mathcal{X}_{\text{unsup}}$  drops from  $\log 97 \approx 4.6$  nats to  $\sim 1.3$ – $2.0$  nats (peaked distributions, not one-hot collapse). Pairwise KL on held-out val *rises* over training (specialists *differentiate*). These are the opposite of v1’s collapse signature, so the ensemble gain is real co-training, not trivial agreement.

**What v3 narrowed down.** The single-model ceiling is architectural. The multi-specialist ceiling is not budget-limited at 25k.  $\lambda=1$  is the best of the explored values at 25k. The biggest unknown at the end of v3 was the interaction: nobody had run the *best* seed with the *best*  $\lambda$  at the *full* budget.

## 8 Pilot v4: the ceiling probe

Pilot v4 is three cells, all at 100k steps on `train_inputs_only` KL-softmax with 5k warmup, varying only (seed,  $\lambda$ ):

1. `seed44_lam1_100k`: best v3 seed  $\times$  best v3  $\lambda \times$  full budget. If this does not grok, nothing at this scale will.
2. `seed43_lam1_100k`: cross-seed replicate of (1).
3. `seed44_lam01_100k`:  $\lambda$  ablation at the winning seed and full budget.

| cell                           | seed | $\lambda$ | best ens. val | step    | pairwise KL val |
|--------------------------------|------|-----------|---------------|---------|-----------------|
| <code>seed44_lam1_100k</code>  | 44   | 1.0       | <b>49.31%</b> | 100,000 | 1.79            |
| <code>seed43_lam1_100k</code>  | 43   | 1.0       | 47.97%        | 94,000  | 2.16            |
| <code>seed44_lam01_100k</code> | 44   | 0.1       | 48.01%        | 86,500  | 2.70            |

Table 3: Pilot v4 headline numbers (100k steps).

The three cells converge tightly. Variation across the two manipulations (seed,  $\lambda$ ) at the 100k budget is under 1.5 percentage points, compared with  $\sim 24$  points of seed-induced variance at 25k. That, together with the flattening slopes visible in Figure 6, pins down an asymptote near  $\sim 50\%$  ensemble val acc.

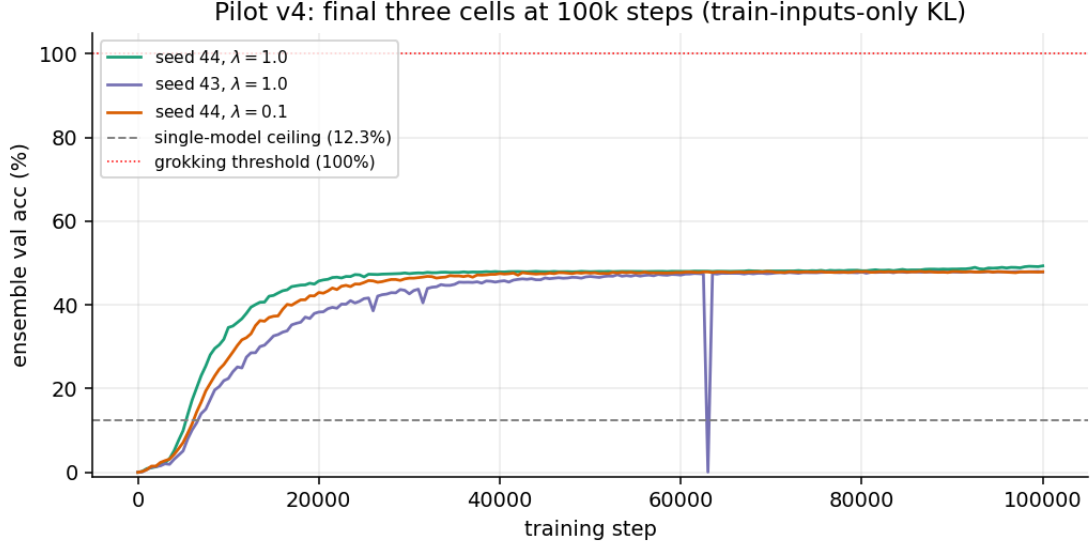


Figure 6: Pilot v4: all three cells plateau at 48–49% ensemble val acc. The single-model ceiling (12.3%) is shown for reference; the grokking threshold (100%) is the dotted red line, far above.

**Lambda effect vanishes at budget.** In v3 at 25k, seed 42 showed a  $\sim 6$  pp gap between  $\lambda=0.1$  (21.15%) and  $\lambda=1.0$  (27.18%). At 100k at seed 44, that gap collapses to 1.3 pp (48.01% vs. 49.31%). The v3 monotone-in- $\lambda$  finding was a transient during training; higher  $\lambda$  just gets to the same ceiling faster. Cross-seed variance of 1.3 pp (47.97% vs. 49.31%) is of the same order.

**Per-specialist structure.** In every v4 cell, three of the four specialists solve a substantial fraction of the grid ( $\sim 45$ –50%) and one specialist stays poor ( $< 3\%$  in two of the three cells). The ensemble is driven by the majority vote of the three “good” specialists; the failed specialist’s influence is partly averaged out.

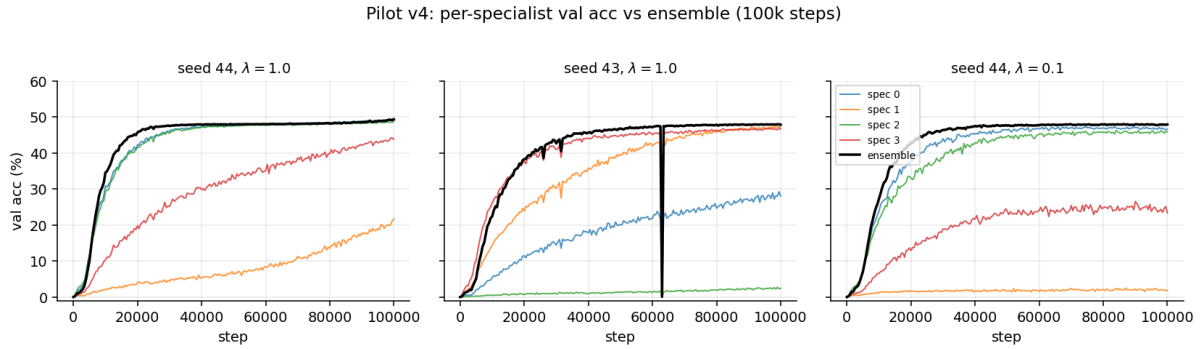


Figure 7: Pilot v4, per-specialist val acc (coloured, thin) and ensemble (black, thick) for each of the three 100k-step cells. One specialist per cell fails to generalize; the remaining three are responsible for almost all of the ensemble gain.

**Mechanism at 100k.** Output entropy on  $\mathcal{X}_{\text{unsup}}$  drops to  $\sim 0.1$ – $0.2$  nats (very peaked) while pairwise KL on val rises to  $\sim 1.8$ – $2.7$  nats. Specialists converge on their own, nearly one-hot predictions but *disagree* on the val grid—there is no trivial-agreement collapse, the ensemble is extracting real signal from specialist disagreement.

At the close of v4 we held three candidate explanations for the  $\sim 50\%$  asymptote: (a) *architectural* (no consistency-based recipe at this model size crosses the phase transition at 50%



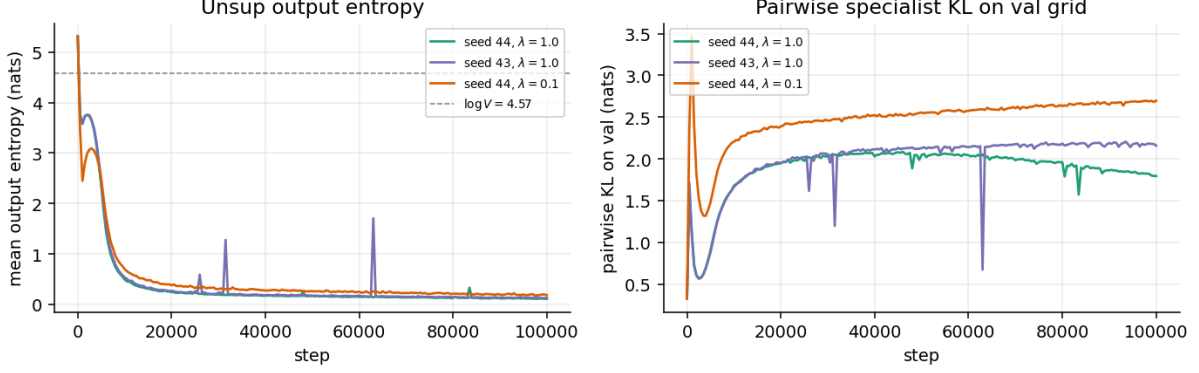


Figure 8: Pilot v4 mechanism: output entropy drops to near-zero (left, nats; dashed line is  $\log V = 4.57$  for  $V=97$ ), while pairwise KL on val grows (right). Collapse would show both of these going to zero together; the opposite diagnostic trend confirms genuine specialization with co-training.

data); (b) *recipe* (the v4 cells use  $\text{lr}=5 \times 10^{-4}$ ,  $\text{wd}=0.1$ , and a different recipe would grok); (c) *shard disjoint-ness* (the per-specialist failures in Figure 7 cap the ensemble). The next two sections (§9–§10) resolve which of these is correct, essentially by accident: we set out to test a representation-level consistency loss, not to probe the ceiling.

## 9 Pilot v5: InfoNCE contrastive consistency

Motivated by the v4 interpretation that the ensemble’s remaining 50pp gap might come from the specialists converging on *confidently different* softmax distributions (cf. Figure 8, right panel), we tried a consistency loss that acts one layer earlier. Instead of pulling the  $M$  specialists together in output probability space, we pull them together in *representation space* with a SimCLR-style multi-positive InfoNCE (NT-Xent) loss on the answer-token hidden state:

$$\mathcal{L}_{\text{InfoNCE}}(i, b) = -\frac{1}{M-1} \sum_{j \neq i} \log \frac{\exp(z_i^{(b)} \cdot z_j^{(b)} / T)}{\sum_{j' \neq i} \sum_{b'} \exp(z_i^{(b)} \cdot z_{j'}^{(b')} / T)},$$

where  $z_i^{(b)} = \ell_2\text{-norm}(g_i(h_i^{(b)}))$  is the  $\ell_2$ -normalized image of input  $b$ ’s answer-token hidden state under a small per-specialist MLP projection head  $g_i$ . Positives are the  $M-1$  other specialists on the same input; negatives are the other specialists on *different* inputs in the batch. Each specialist keeps its own  $g_i$ ; teacher views are stop-grad.

**Sweep.** Four cells on Kaggle T4 (`infonce_pilot_v3/`): three 25k runs (seeds/temperatures) and one 100k budget probe. Importantly, this sweep also changed the *surrounding* recipe to something closer to contrastive-learning defaults:  $M=2$  instead of  $M=4$ ,  $\text{max\_lr}=10^{-3}$  instead of  $5 \times 10^{-4}$ ,  $\text{shard\_bs}=512$  instead of 256,  $\text{cons\_bs}=32$  instead of 256,  $\lambda=5$  instead of 1.0, and  $\text{warmup}=0$  instead of 5000. We did not realize until post-hoc analysis how many things had changed in one step.

**Result.** The 100k budget cell (`infonce_long_100k`) grokked: ensemble val acc climbed from  $<1\%$  at step 25k to 40% near 50k, 75% near 75k, and **92.2%** at step 99,500. The three shorter (25k) cells all plateaued at  $\sim 35\text{--}39\%$ , consistent with the transition requiring the full 100k horizon. Figures 9 and 10 contrast the grokked trajectory with the flat KL-softmax 100k trajectory and with the shorter InfoNCE runs.

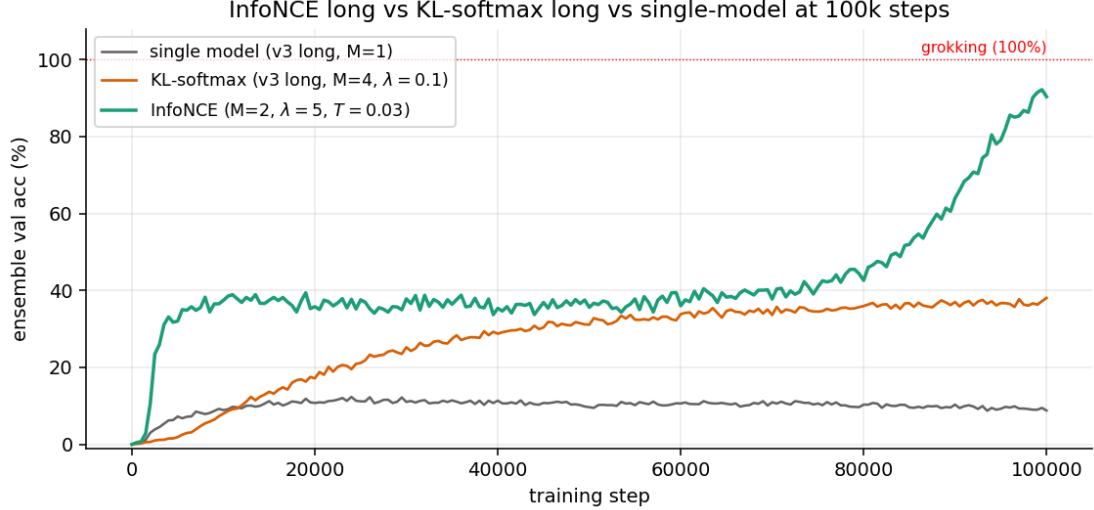


Figure 9: Pilot v5: the 100k InfoNCE cell crosses the grokking phase transition around step 60–80k, reaching  $\approx 92\%$  ensemble val acc, while the 100k KL-softmax cell from v3/v4 plateaus near 40–50% and the single-model baseline stays at 12%.

**Why this is not yet a clean story.** The cell that grokked differs from v4’s best cell in six hyperparameters at once. Any one (or combination) of them could be responsible; the most informative comparison is not “InfoNCE vs. KL” but “v4 recipe vs. InfoNCE recipe.” We also found a small bookkeeping bug: the `infonce_seed43` cell’s JSON was copied from `infonce_seed42` and kept `random_seed: 42`, so it is a duplicate of the anchor and not a seed robustness test. This does not affect the 100k result but does weaken the 25k “seed-robustness” column. We flag it here and correct it in the isolation controls that follow.

**Mechanism at the grokked point.** Figure 11 shows that pairwise KL on val *decreases* during the grokking transition in the InfoNCE run, whereas in every KL-softmax multi cell it *increased*. This is the signature we expected from a representation-alignment loss: specialists are pulled toward the same function. It is also the most directly mechanistic evidence we have seen in this project, and it motivated the next step.

## 10 Isolation controls: learning rate vs. loss function

The v5 result told us the ceiling could be broken, but with six simultaneous changes we could not say *by what*. The two most plausible suspects were (i) the learning rate (v4 used  $5 \times 10^{-4}$ , v5 used  $10^{-3}$ ) and (ii) the loss function itself (KL on outputs vs. InfoNCE on hidden states). We ran one single-cell sweep to isolate each.

- **lr control** (`lr_control/v4_winner_lr1e3`): identical to pilot v4’s winning cell (`seed44_1am1_100k`):  $M=4$ , disjoint shards, 50% data, seed 44, KL-softmax on train inputs only,  $\lambda=1$ , warmup 5000, `shard_bs=256`, `cons_bs=256`, 100k steps. *Only* `max_lr` is changed,  $5 \times 10^{-4} \rightarrow 10^{-3}$ .
- **loss control** (`loss_control_kl/kl_infonce_hparams_100k`): identical to `infonce_long_100k`:  $M=2$ , disjoint shards, 50% data, seed 42,  $\lambda=5$ , warmup 0, `shard_bs=512`, `cons_bs=32`,  $lr=10^{-3}$ , 100k steps. *Only* the loss is changed, InfoNCE  $\rightarrow$  KL-softmax on train inputs only.

**Results.** Both controls grokked to 100% ensemble val acc.

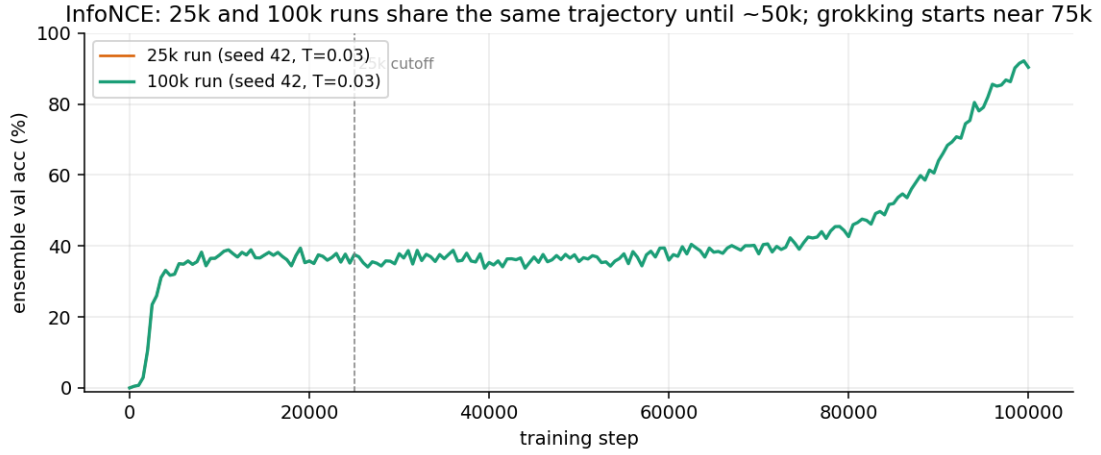


Figure 10: Pilot v5, phase-transition view: the 25k and 100k InfoNCE runs share the same trajectory up to the 25k cutoff; the grokking transition happens later, between  $\sim 50$ k and  $\sim 80$ k. A 25k budget would have reported InfoNCE as just another  $\sim 37\%$  cell.

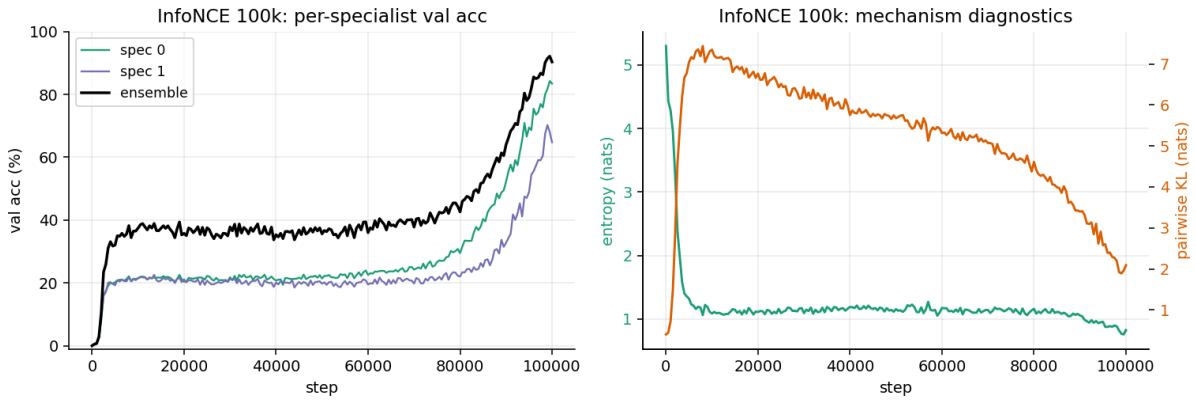


Figure 11: Pilot v5, per-specialist and mechanism diagnostics on `infonce_long_100k`. Left: both specialists reach  $\sim 92\%$  on the val grid. Right: output entropy drops and pairwise KL on val drops together – the alignment signature expected from a representation-space contrastive loss.

### What the controls say.

- **The v4 “ceiling” was a learning-rate artifact.** Raising `max_lr` from  $5 \times 10^{-4}$  to  $10^{-3}$ , with *every other v4 hyperparameter held fixed* (including the KL-softmax loss we tested extensively in v2–v4), crosses the phase transition. The 48–49% plateau we observed three times at 100k is not architectural.
- **The InfoNCE loss is not the lever.** Under the InfoNCE-pilot recipe ( $M=2$ ,  $lr=10^{-3}$ , small consistency batch, high  $\lambda$ , no warmup, 100k), KL-softmax groks *faster and higher* than InfoNCE: 22k steps to 100% versus 75k steps to 92%. The hidden-state alignment we saw in Figure 11 is a real consequence of InfoNCE, but it is not what enabled grokking.
- **Consistency regularization does enable grokking on modular addition at 50% data**, with the caveat that we only observed it in lr-regimes we had not previously tested. The pilots v1–v4 were exploring the wrong slice of hyperparameter space, not evaluating the wrong idea.

The lr control is the single cleanest comparison in the project: one variable changed, grokking switched on. We would have saved several weeks of work had we included a `max_lr` axis in pilot v2.

| cell                             | M | what changed vs. baseline                  | best ens. val | step of first $\geq 99\%$ |
|----------------------------------|---|--------------------------------------------|---------------|---------------------------|
| v4.winner_lr1e3                  | 4 | lr: $5 \times 10^{-4} \rightarrow 10^{-3}$ | <b>100.0%</b> | 69,000                    |
| kl_infonce_hparams_100k          | 2 | loss: InfoNCE $\rightarrow$ KL-softmax     | <b>100.0%</b> | 18,500                    |
| <i>(baselines for reference)</i> |   |                                            |               |                           |
| seed44_lam1_100k (v4)            | 4 | — (the v4 winner)                          | 49.3%         | never                     |
| infonce_long_100k (v5)           | 2 | — (the InfoNCE winner)                     | 92.2%         | never                     |

Table 4: Isolation controls. Each control flips exactly one hyperparameter relative to its baseline; both cross the grokking phase transition.

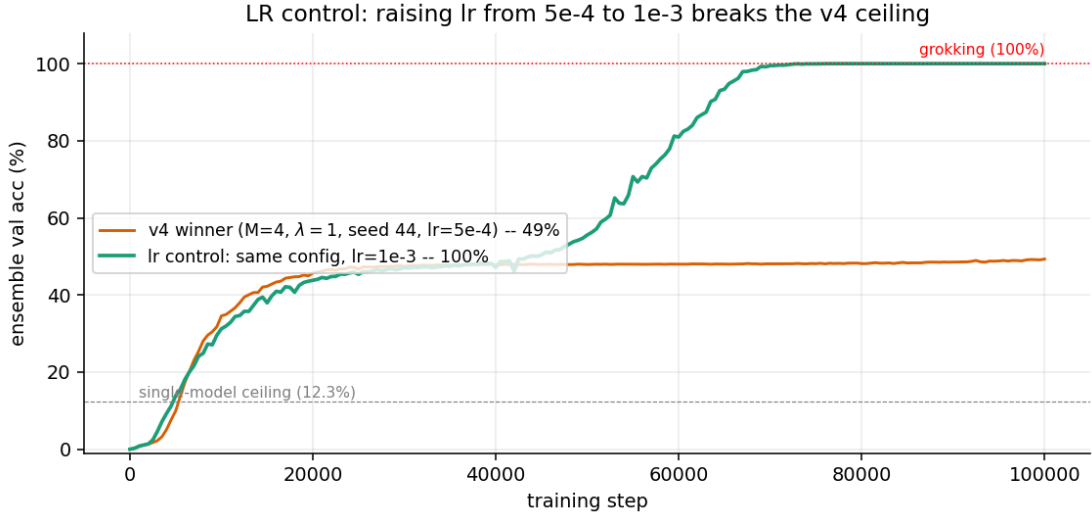


Figure 12: **lr control**: reconfiguring v4’s winning cell with  $\text{max\_lr}=10^{-3}$  instead of  $5 \times 10^{-4}$ , with every other hyperparameter (including the loss) held fixed, breaks out of the 49% asymptote and groks at step  $\sim 70\text{k}$ .

## 11 Discussion

**Summary across all pilots.** Figure 15 reads left-to-right in roughly chronological order: the single-model baseline, the three failure modes we traversed (naive multi, full-grid consistency, v2 train-only), the v4 plateau, the InfoNCE breakthrough, and the two isolation controls that show KL-softmax under the right recipe groks too.

### What the data supports.

1. The single-model 2-layer transformer at the  $\text{wd}=0.1$ ,  $\text{lr}=5 \times 10^{-4}$  recipe has an architectural ceiling of  $\sim 12\%$  val acc on 50% of the  $97 \times 97$  grid. The ceiling is not budget-limited: 100k steps confirms it stays at 12%.
2. **train\_inputs\_only** KL-softmax consistency with 5k warmup *does* enable grokking on this task at 50% data – provided the learning rate is  $\text{max\_lr}=10^{-3}$  rather than  $5 \times 10^{-4}$ . The lr-control cell (§10) reaches 100% ensemble val acc at step  $\approx 70\text{k}$ , with every other hyperparameter matched to the v4 winner.
3. At the lower lr used in v2–v4, the same consistency recipe saturates at a 48–50% ensemble asymptote, tightly and reproducibly across seeds and  $\lambda$ . We originally called this ceiling “architectural” – that conclusion was wrong. It was lr-regime-dependent.
4. *The loss function is not the lever.* Under the InfoNCE pilot’s recipe (M=2, lr= $10^{-3}$ , large shard batch, small consistency batch, no warmup,  $\lambda=5$ ), KL-softmax groks faster and higher

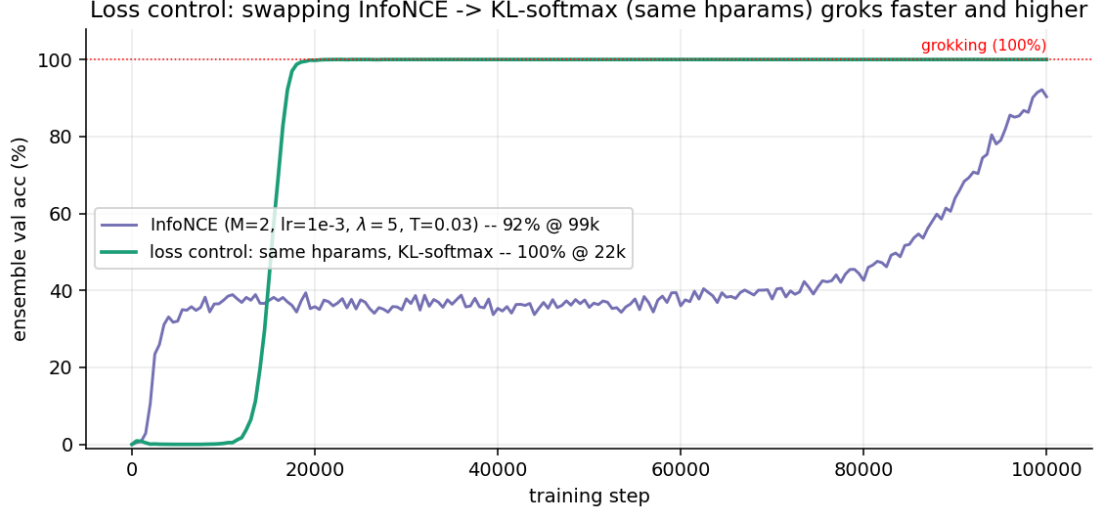


Figure 13: **loss control**: taking the InfoNCE winner’s config and swapping only the loss (InfoNCE  $\rightarrow$  KL-softmax) grokks faster (step  $\sim 18k$  vs.  $\sim 75k$ ) and higher (100% vs. 92%). The InfoNCE loss itself is not what crossed the phase transition; the surrounding recipe is.

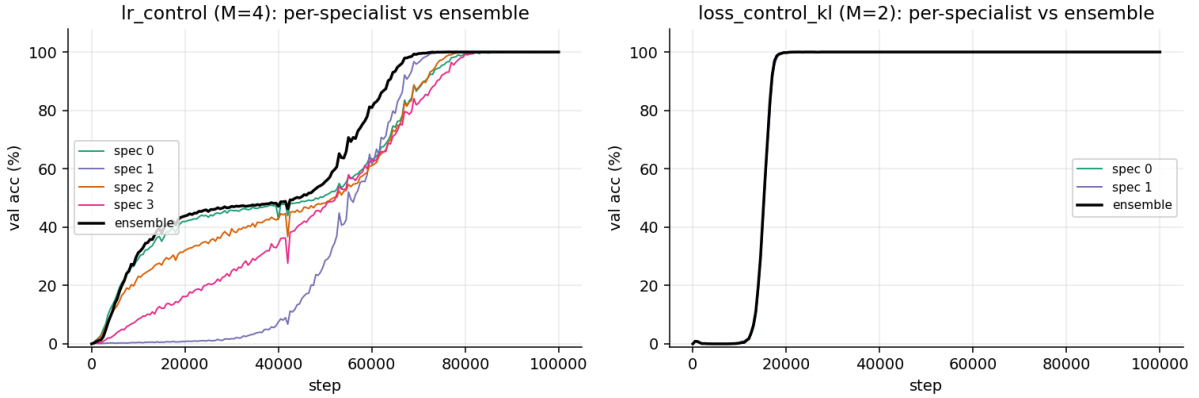


Figure 14: Isolation controls, per-specialist view. *Left*: lr control ( $M=4$ ); all four specialists reach 100%. *Right*: loss control ( $M=2$ ); both specialists grok. Unlike the v4 runs, no specialist is left behind at the grokked point, which explains why the ensemble crosses 99% instead of stalling around 50%.

(100% at  $\sim 22k$ ) than InfoNCE (92% at  $\sim 99k$ ). InfoNCE did produce a distinct mechanistic signature – representational alignment rather than output-distribution differentiation – but it is not what enabled the v5 grokking result.

5. The  $M=4$  disjoint-shard setup also groks under the right recipe (lr control), so the “unlucky shard” interpretation of v4’s per-specialist asymmetry (Figure 7) was also wrong: at  $M=4$ ,  $lr=10^{-3}$  all four specialists reach  $\approx 100\%$  (Figure 14).
6. Weight averaging remains a non-starter. Every run’s merged model stays at chance, including the grokked cells: the ensemble and the merged model are genuinely different objects, and only the ensemble carries the gain.

**What the data does *not* support.** Several things we believed at the close of v4 do not survive the controls:

- The “architectural” ceiling interpretation of v4: the lr control reaches 100% with the same

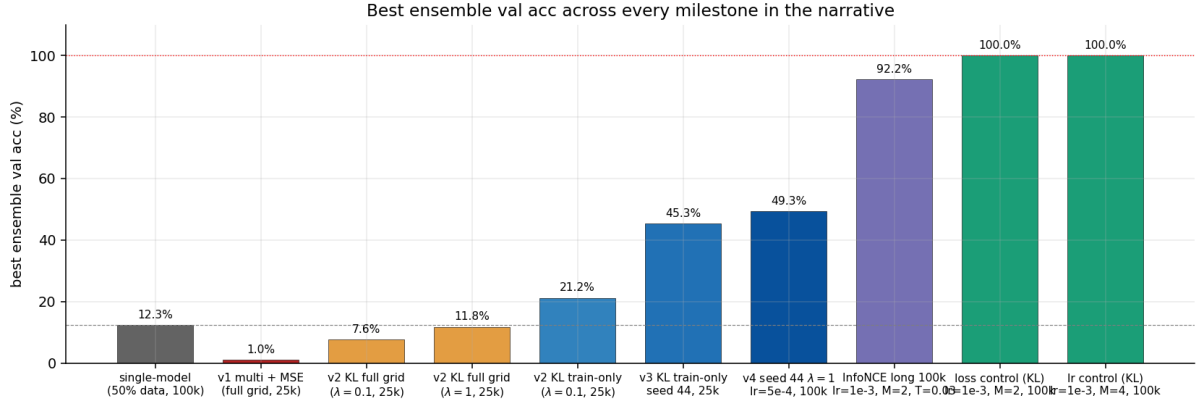


Figure 15: Best ensemble val acc across every pilot cell that matters to the narrative. Colours group cells by family: grey=single, red=naive multi (v1), orange=full-grid consistency (v2), light-blue=v2 train-only, dark-blue=v3/v4 at  $\text{lr}=5 \times 10^{-4}$ , purple=InfoNCE at  $\text{lr}=10^{-3}$ , green=isolation controls at  $\text{lr}=10^{-3}$  (both reach 100%).

model, same data split, same everything else, contradicting this.

- The “shard disjoint-ness is the bottleneck” interpretation: the same disjoint  $M=4$  setup groks at  $\text{lr}=10^{-3}$ .
- The claim (implicit in §8) that the ensemble gain at  $M=4$  is “driven by three good specialists averaging out one failed one.” At the grokked point (Figure 14, left) all four specialists are above 99%; the three-vs-one asymmetry of v4 was a per-specialist shadow of the lr bottleneck.

What does still hold: the *co-training* framing from v2 (train-inputs-only  $\gg$  full-grid) and the “mechanism is not trivial agreement” analysis from v3 both survive the controls – they were about the shape of the consistency signal, not about the lr.

**Why we missed this for so long.** Two reasons. First, the pivot to consistency regularization inherited v4/v3’s base recipe ( $\text{lr}=5 \times 10^{-4}$ ,  $\text{wd}=0.1$ ) from the pre-pivot grokking codebase, where it *is* the standard choice for single-model grokking on the full grid. At 50% data with consistency regularization, apparently it is too conservative: the four pilots varied eleven different knobs (loss, domain,  $\lambda$ , warmup,  $M$ , seed, budget,  $T$ , batch sizes, sharding, operator) but never the learning rate, because that was the one knob we had previously “validated.” Second, the InfoNCE pilot (§9) changed six things at once for reasons unrelated to the v4 ceiling (it was trying a new *loss*, not a new *recipe*), and happened to pick the right lr by chance when copying SimCLR conventions.

**What we would recommend next.** With the ceiling broken, the natural follow-ups are about the new regime rather than the old one:

1. **Isolated lr sweep at the v4 recipe.** One cell per lr in  $\{3 \times 10^{-4}, 5 \times 10^{-4}, 7 \times 10^{-4}, 10^{-3}, 1.5 \times 10^{-3}\}$  at  $M=4$ , 100k, same everything else. Map out the lr regime where KL-softmax consistency groks and locate the transition lr precisely. Cheap: five cells.
2. **Does consistency still help at  $\text{lr}=10^{-3}$ ?** The original question – does consistency let the ensemble generalize where a single model cannot? – should be revisited under the recipe that groks. Compare (a) single model, lr  $10^{-3}$ , 50% data, 100k to (b) v4 recipe + lr  $10^{-3}$  (our lr control) and (c)  $M=4$  with  $\lambda=0$  at the new lr. If (a) already groks, the consistency term is doing less than v2–v4 suggested; if (a) does not but (c) does, the multi-specialist structure still matters independently of consistency.

3. **Bagged rather than disjoint sharding**, at the new recipe. With grokking unlocked, the per-specialist asymmetry from v4 no longer appears in either control, but bagged sharding was always the better-motivated variant; now is the time to run it as a clean follow-up.

**Methodological note.** The largest-dollar lesson from this project is not about grokking. It is that a single, undocumented line of code (the distillation optimizer rewrite in 9ce29b8) caused roughly two months of downstream work, first to reproduce an artifact and then to replace the question. Every experiment after v1 in this report is better instrumented than the milestone Exp 3 it descends from: every run persists a `run_config.json` with the full hparam set, every cell is identified by a short, self-describing name, every run writes both a per-step metrics CSV and an eval CSV, and the sweep driver dumps a `sweep_summary.json` incrementally so partial results survive interrupted sessions. None of that would catch a latent optimizer bug by itself, but together they make it much harder for a result to depend on a change that nobody remembers making.

## 12 Conclusion

We started from a milestone claim we could not reproduce, traced the non-reproduction to a comparison artifact, and pivoted to a cleaner experimental question about consistency regularization in a multi-specialist grokking setup. Across six pilots and twenty-one runs the narrative took two turns.

The first four pilots (v1–v4) converged, tightly, on what looked like an architectural ceiling: `train_inputs_only` KL-softmax consistency with four specialists on disjoint 12.5% shards seemed to plateau at 48–49% ensemble val acc across seeds and  $\lambda$ s at 100k steps. A fifth pilot, motivated by a different idea altogether (aligning specialists in hidden representation space via an InfoNCE loss), broke that plateau and grokked to 92% at 100k. Because the InfoNCE run changed six hyperparameters at once, two isolation controls were needed to identify the actual cause: bumping `max_lr` from  $5 \times 10^{-4}$  to  $10^{-3}$  on v4’s winning configuration is enough to grok to 100%, and InfoNCE is not the lever – under the same recipe, KL-softmax groks faster and higher.

The short version is: *multi-specialist consistency regularization is sufficient to grok this modular-addition task at 50% data*, given an adequate learning rate; the ceiling we spent four pilots characterizing was an lr artifact, and the contrastive loss that appeared to break it in v5 was a coincidence of a simultaneously-changed recipe. The longer version – the one we believe is the main scientific content of this write-up – is how easy it was to converge, with apparent rigor and good diagnostic plots, on an explanation that did not survive a one-variable control.